Reflection Questions

1. **How did Pylint contribute to code maintainability?** Pylint enforced clean, readable coding standards by flagging snake_case violations, detecting missing docstrings, and identifying unused imports (os, sys, json) and unused variables. This makes the codebase easier for future developers to read and maintain.

2. **How did Bandit improve code security?** Bandit proactively caught critical vulnerabilities before deployment, specifically flagging a weak 1024-bit RSA key, the insecure ECB cipher mode, and XML parsing vulnerabilities linked to minidom.

3. **What is the value of Flake8 in formatting?** Flake8 acted as a strict grammar checker, standardizing PEP 8 guidelines by highlighting missing blank lines, lines exceeding 79 characters, and overly broad except: clauses.

4. **Overall impact on quality?** Using all three tools concurrently allowed for a layered review process—formatting, logic/maintainability, and security—drastically reducing the likelihood of runtime bugs and breaches.